

Treinamento em Biologia Computacional - Módulo 2

Equipe de Dados Biológicos (EDB)

Grupo de Design de Proteínas (GDP)

Laboratório de Biologia Computacional (LBC)

Dia 2 (07/11/2025)



Módulo 2

Visualização molecular de biomoléculas no ChimeraX

Equipe de Dados Biológicos (EDB)

João V. S. Guerra

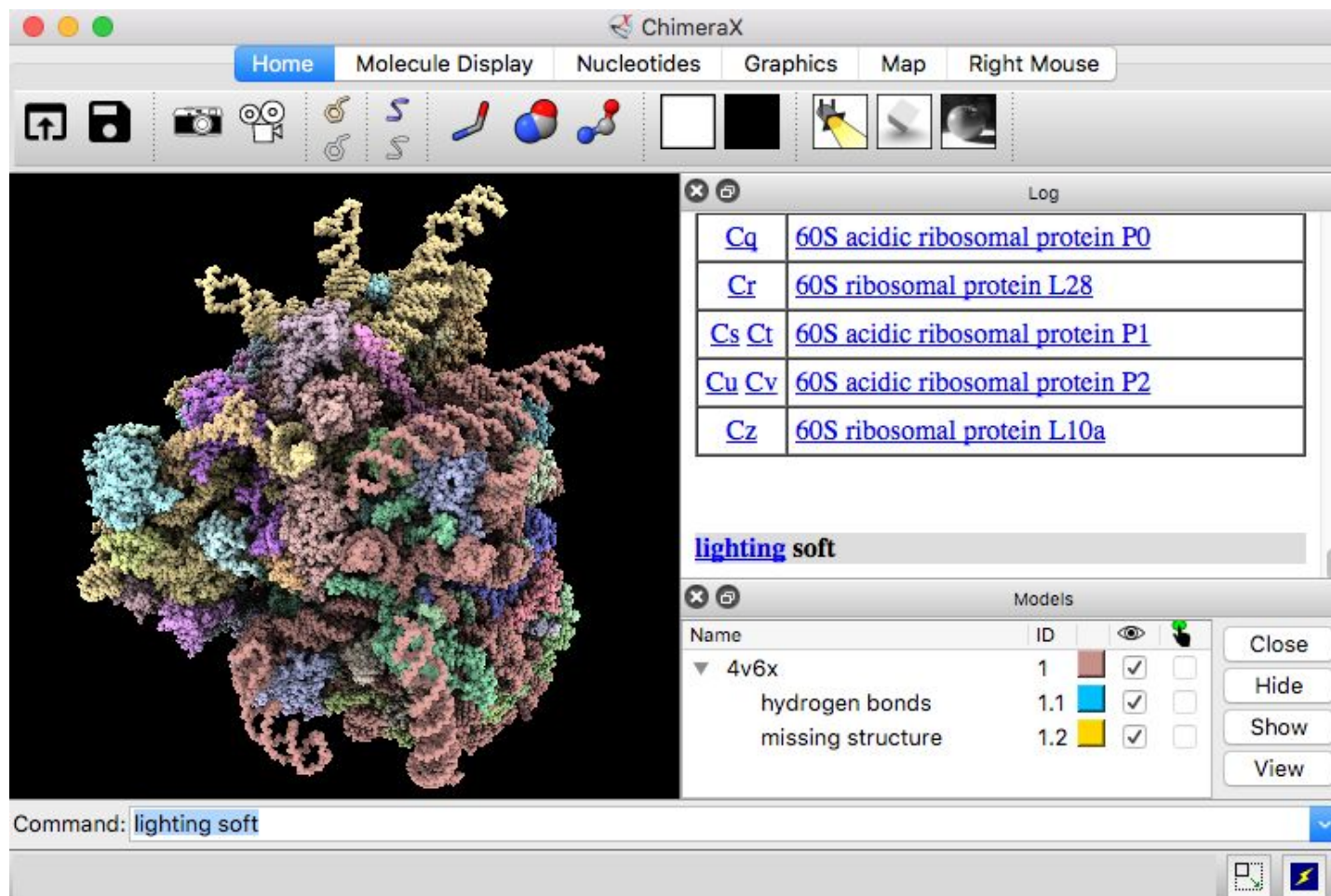
Grupo de Design de Proteínas (GDP)

Helder V. Ribeiro-Filho

Dia 2 (07/11/2025)



- » O UCSF [ChimeraX](https://www.cgl.ucsf.edu/chimerax) é um software de código aberto de última geração para visualização, análise e manipulação de estruturas moleculares e dados relacionados.

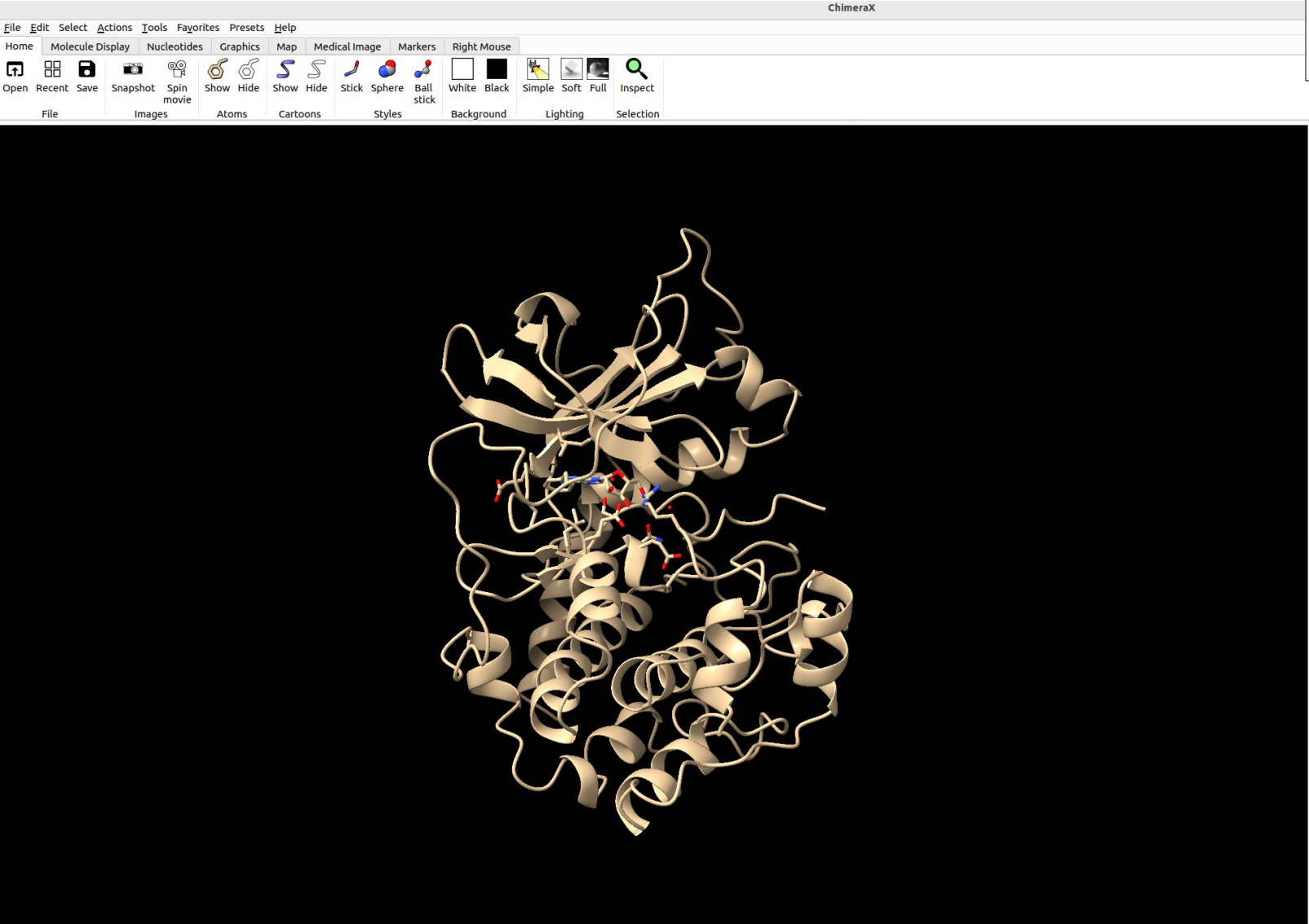


Principais funcionalidades:

- Visualização interativa de moléculas
- Suporte a múltiplos formatos (PDB, mmCIF, MRC etc.)
- Medições (distâncias, ângulos, contatos) e manipulação estrutural
- Análise de mapas de crio-EM e dados volumétricos
- Integração com bancos de dados (PDB, AlphaFold, UniProt)
- Geração de imagens e animações de alta qualidade
- Suporte a scripts em Python e plugins

Download: www.cgl.ucsf.edu/chimerax

A interface de visualização do ChimeraX



Show Sequence from Structure

Usage: sequence chain *chain-spec* [viewer true | false]

Log

Startup Messages

note

available bundle cache has not been initialized yet

UCSF ChimeraX version: 1.9.dev202410230938 (2024-10-23)
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[How to cite UCSF ChimeraX](#)
open 1fmo format mmCIF fromDatabase pdb
1fmo title:
Crystal structure of A polyhistidine-tagged recombinant catalytic subunit of camp-dependent protein kinase complexed with the peptide inhibitor pki(5-24) and adenosine
[\[more info...\]](#)

Chain information for 1fmo #1		
Chain	Description	UniProt
E	CAMP-DEPENDENT PROTEIN KINASE	KAPCA_MOUSE 1-350
I	HEAT STABLE RABBIT SKELETAL MUSCLE INHIBITOR PROTEIN	IPKA_RABIT 5-24

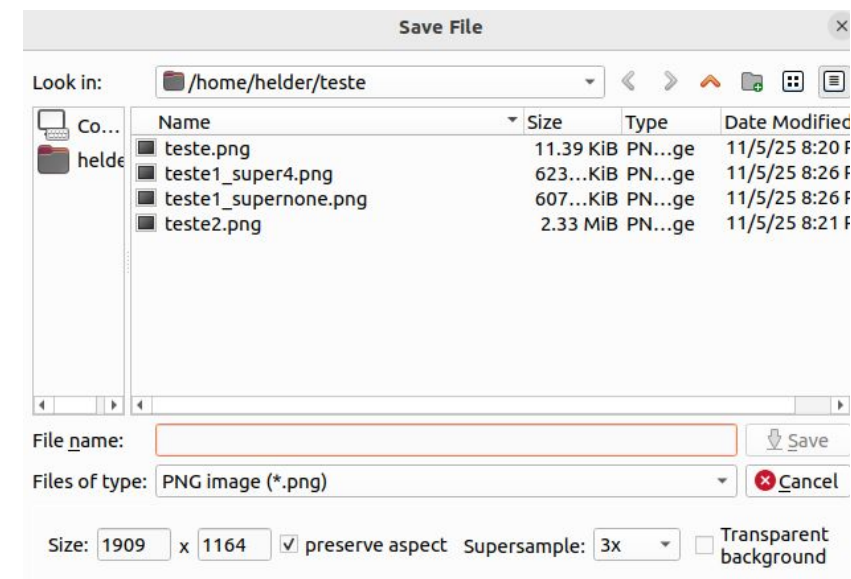
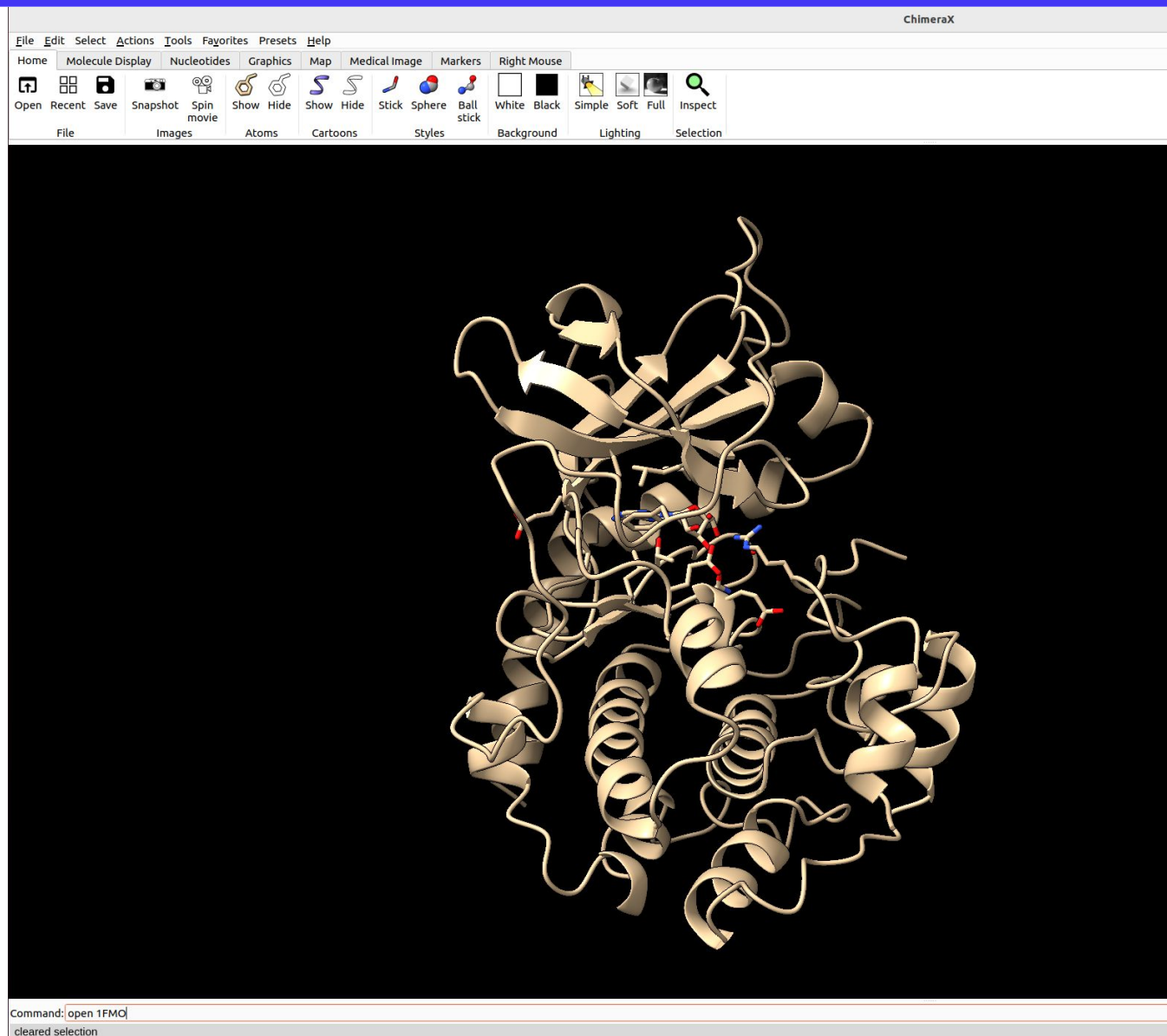
Non-standard residues in 1fmo #1

[ADN — adenosine](#)

Models

Name	ID			Skip	
1fmo	1		☒	☐	Close Hide Show View Info

Abrindo e salvando uma estrutura/seção no ChimeraX



Hierarchical Specifiers

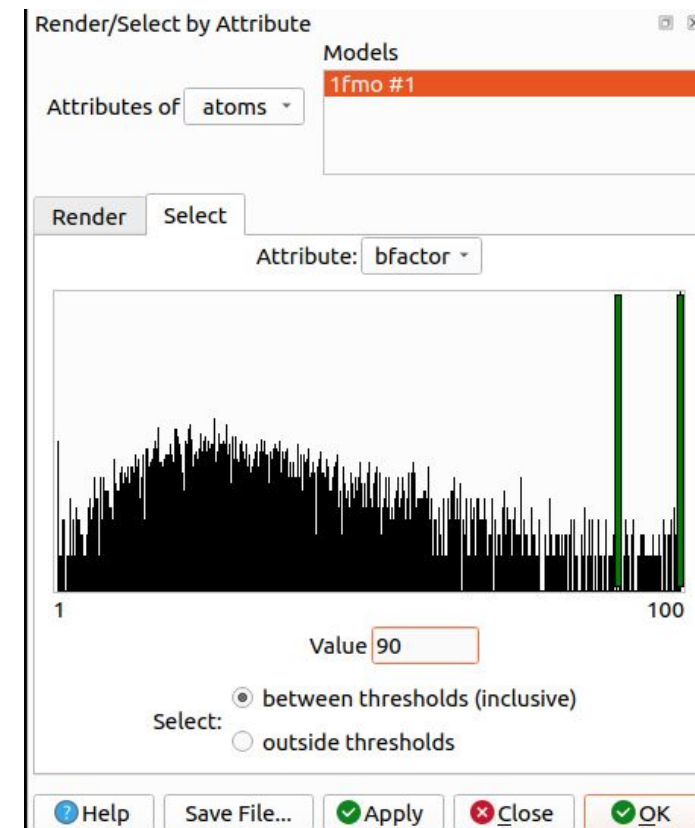
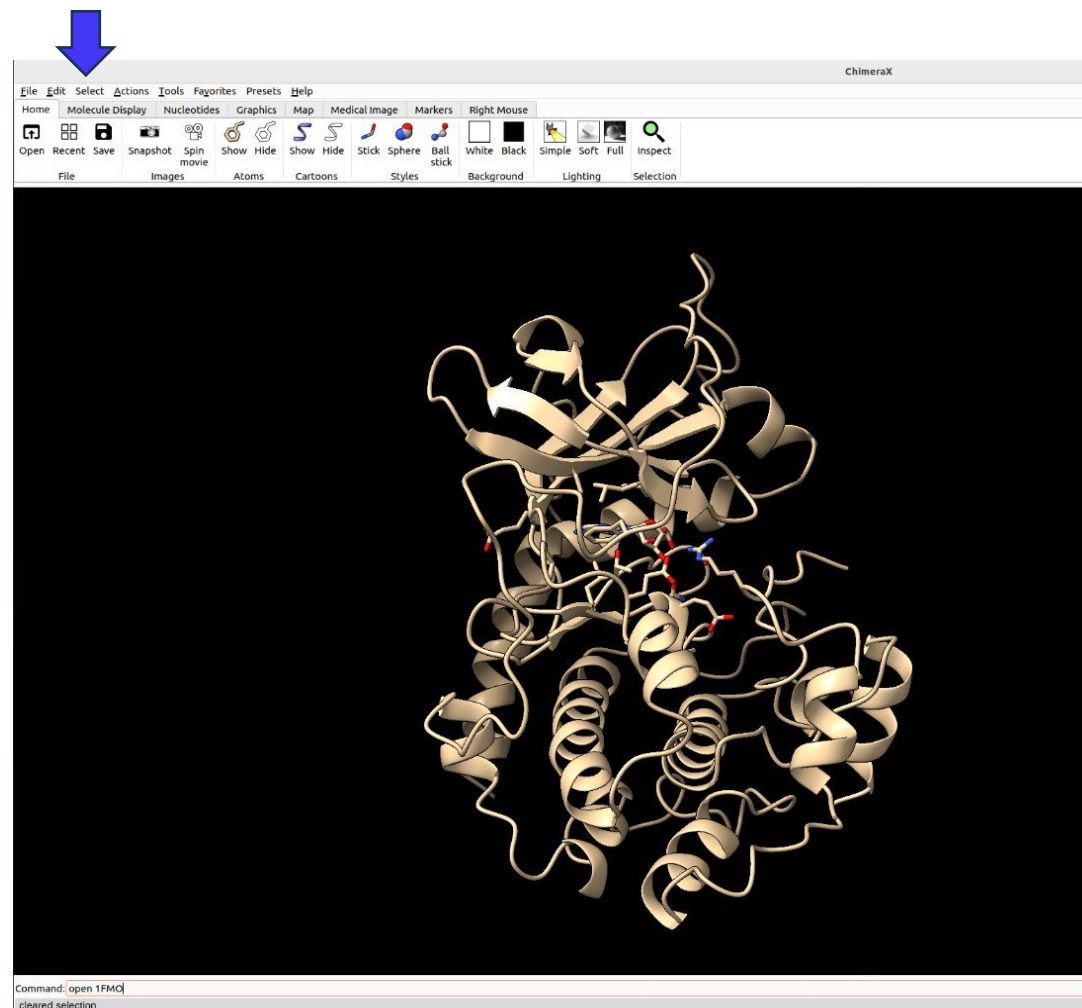
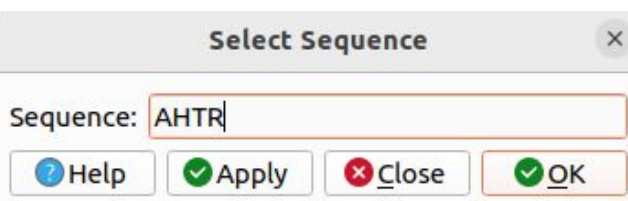
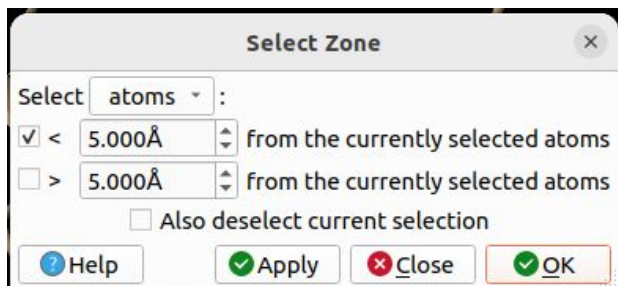
Symbol	Reference Level	Definition	Examples
#	model	model number assigned to the data in ChimeraX (hierarchical, with positive integers separated by dots: N, N.N, N.N.N, <i>etc.</i>)	#1 #1.3
/	chain	chain identifier (case-insensitive unless both upper- and lowercase chain IDs are present)	/A
:	residue	residue number <i>OR</i> residue name (case-insensitive)	:51 :glu
@	atom	atom name (case-insensitive)	@ca

<https://www.cgl.ucsf.edu/chimerax/docs/user/commands/atomspec.html#hierarchy>

Seleções no ChimeraX

Principais tipos de seleção:

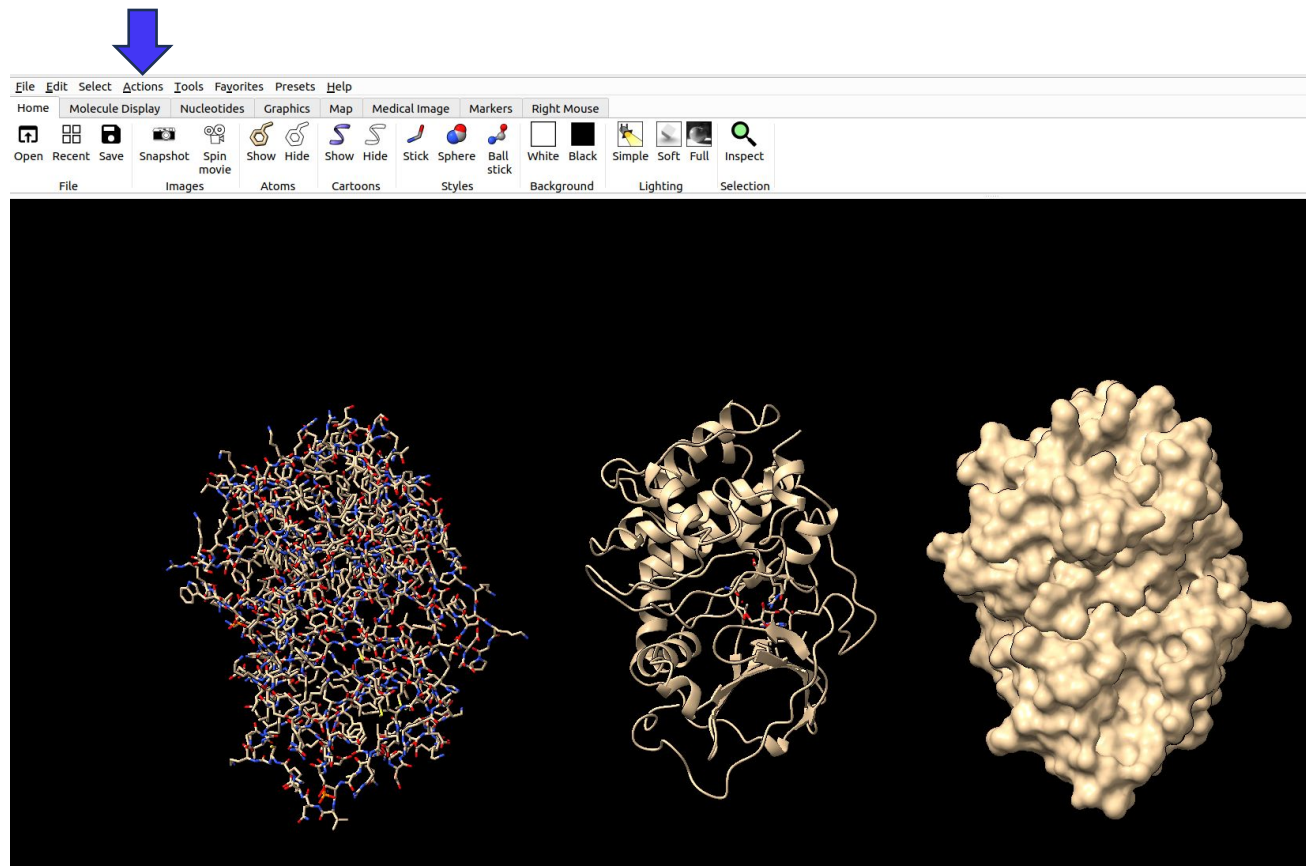
- Cadeia
- Resíduos
- Átomo
- Zona
- Sequência
- Atributo



Ações no ChimeraX

Principais tipos de ações:

- Alterar representações
- Funções *Delete*
- Cores e *labels*
- Funções *Focus* e

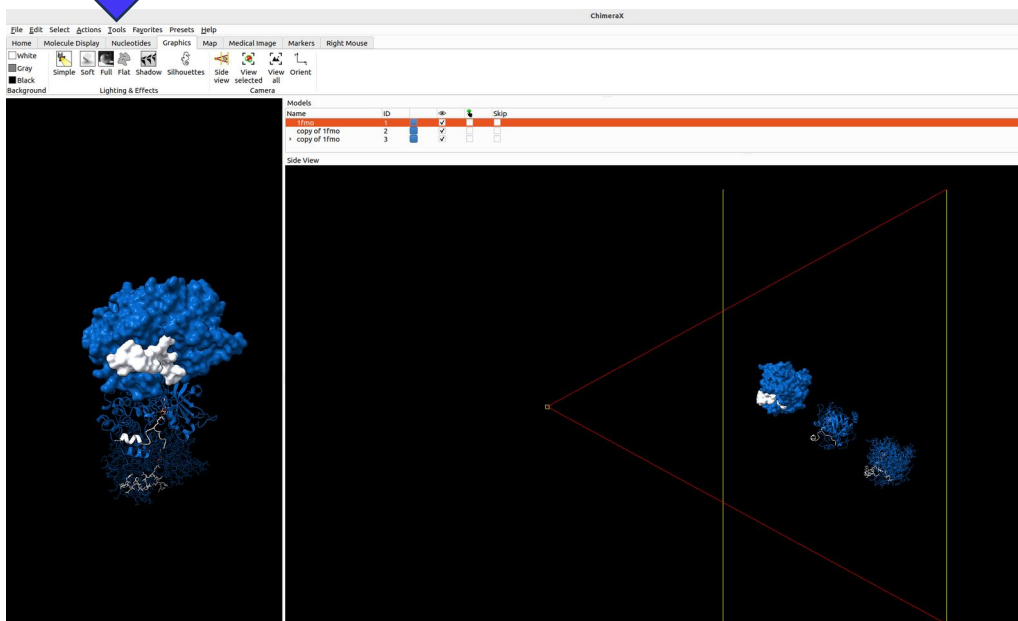


rainbow polymer palette Blues-5
color bychain

<https://www.cgl.ucsf.edu/chimerax/docs/user/commands/palettes.html>

Principais ferramentas:

- Gerais: side view
- Sequência: Visualizador e Alinhamento



CAMP-DEPENDENT PROTEIN KINASE (1fmo (#1) Chain E) [ID: 1/E]

```

chain E 1 GNAAA A K K G S E Q E S V K E F L A K A K E D F L K K W E T P S Q N T A Q L D Q F D R I K T L G T G S F G
chain E 56 R V M L V K H K E S G N H Y A M K I L D K Q K V V K L K Q I E H T L N E K R I L Q A V N F P F L V K L E F S F
chain E 111 K D N S N L Y M V M E Y V A G G E M F S H L R R I G R F S E P H A R F Y A A Q I V L T F E Y L H S L D L I Y R
chain E 166 D L K P E N L L I D Q Q G Y I Q V T D F G F A K R V K G R T W T L C G T P E Y L A P E I I L S K G Y N K A V D
chain E 221 W W A L G V L I Y E M A A G Y P P F A D Q P I Q I Y E K I V S G K V R F P S H F S S D L K D L L R N L L Q V
chain E 276 D L T K R F G N L K N G V N D I K N H K W F A T T D W I A I Y Q R K V E A P F I P K F K G P G D T S N F D D Y
chain E 331 E E E E I R V S I N E K C G K E F T E F
    
```

Clustal Omega Alignment [ID: 2]

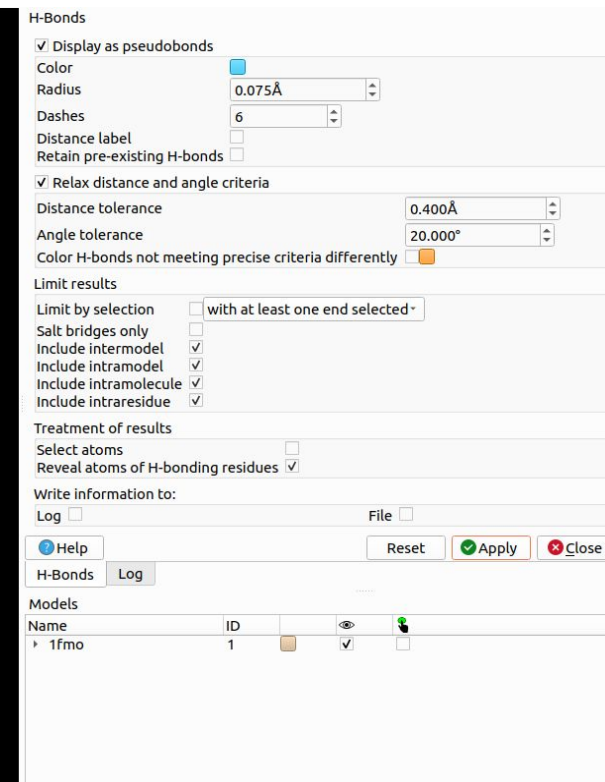
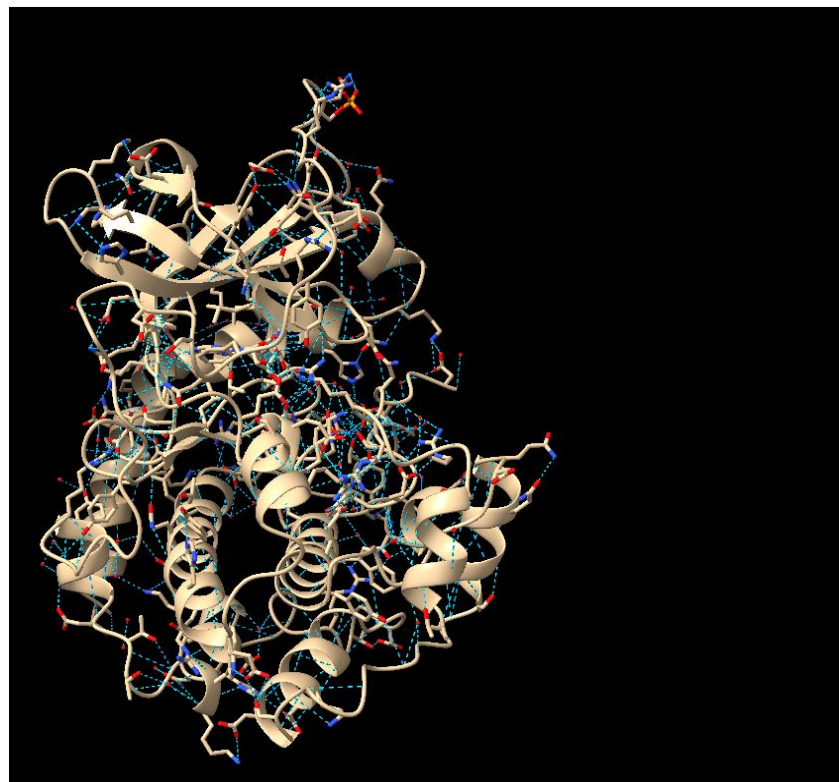
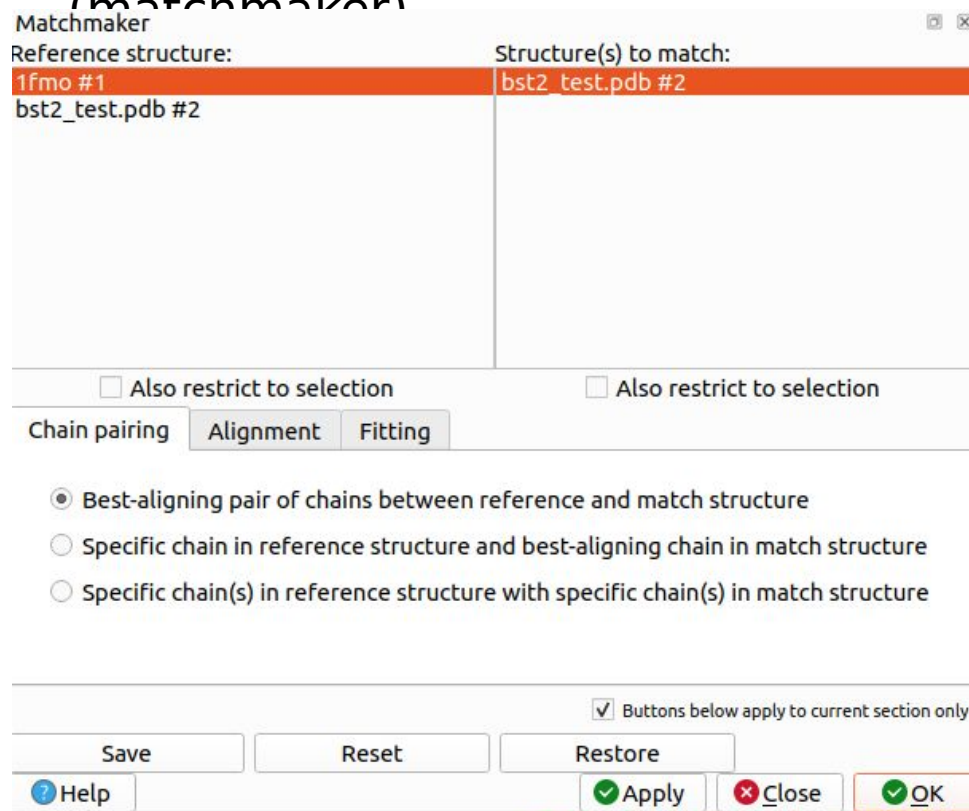
```

Chain A ----- R A F W T E Q E G F E T W D G E T R K V K A N S Q T H R V D E G
Conservation 226          236          246          256          266
chain E ----- K G Y N K A V D W W A L G V L I Y E M A A G Y P P F
chain A T L R G Y Y N Q S E A G S H T V Q R M Y G C D V G S D W R F L ----- R G Y H Q Y
Conservation 271          281          291          301          311
chain E F A D Q P I Q I Y E K I V S G K V R F P S H F S S D L K D L L R N L L Q V D L T K R F G N
chain A A Y D G K D ----- Y I A L K E D L R S W T A A D M A -----
    
```

sequence align
#1/A#2/A

Principais ferramentas:

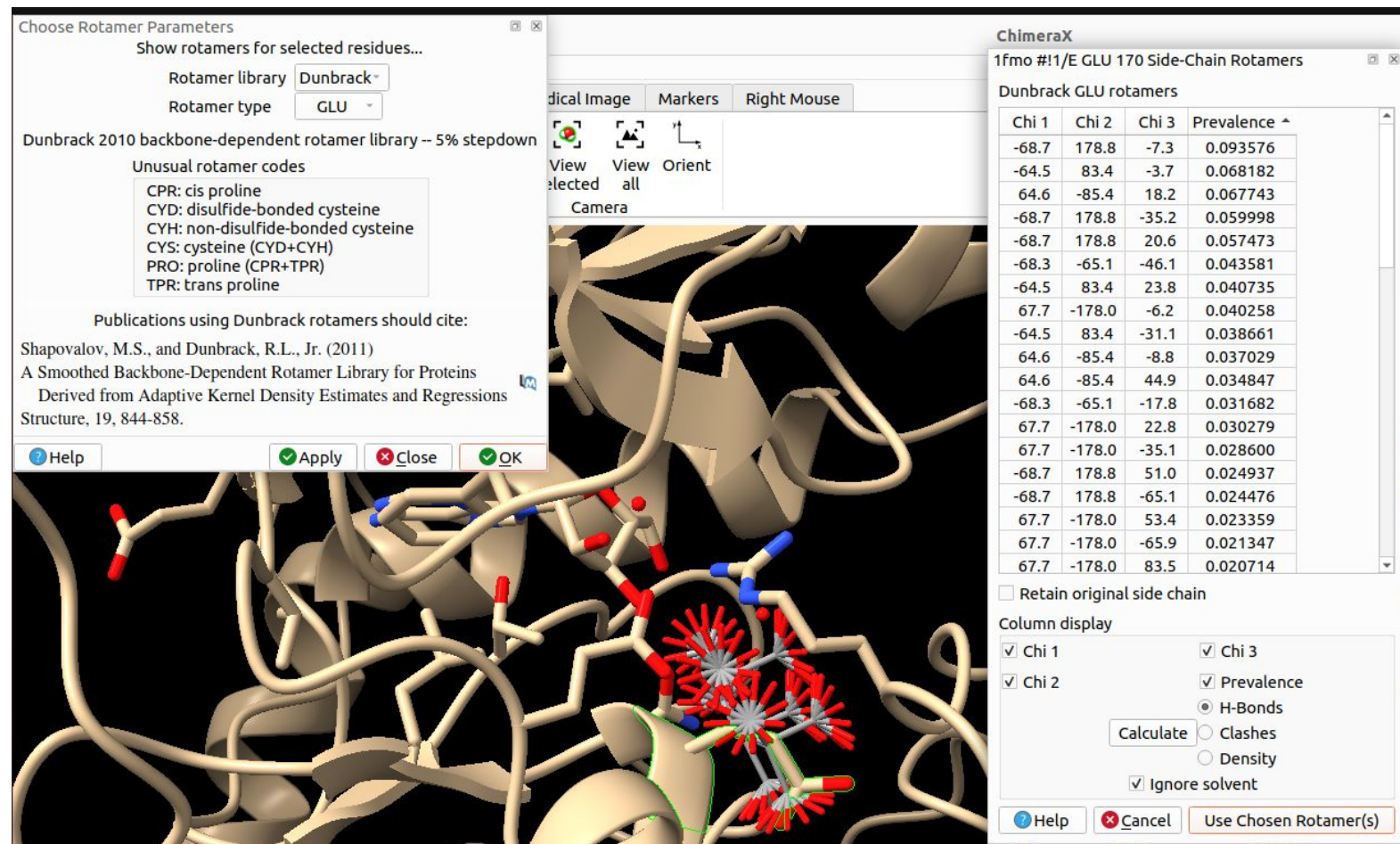
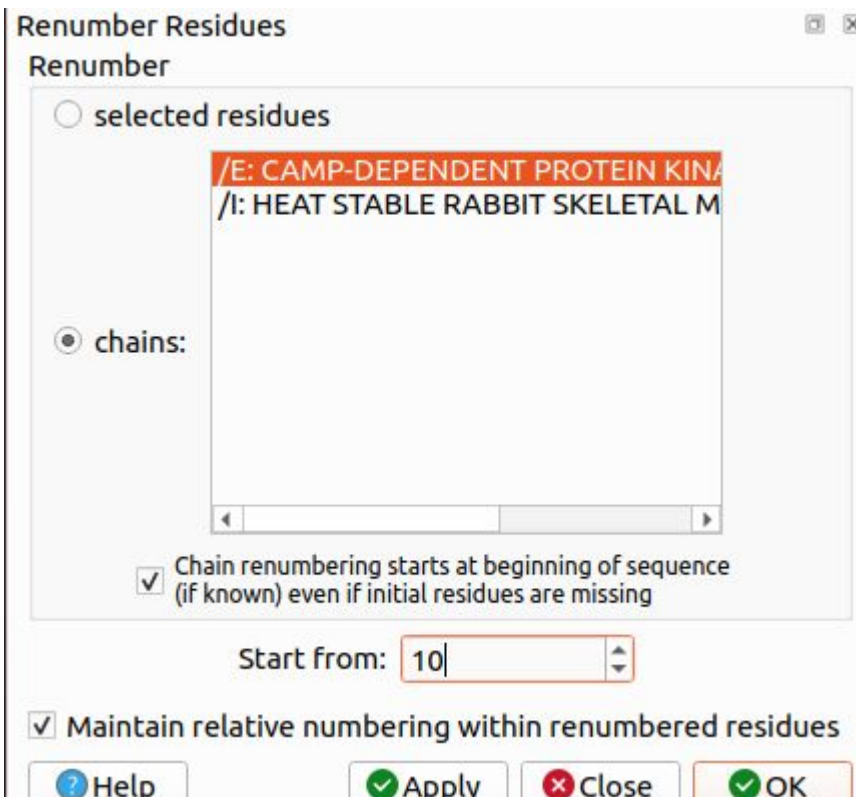
- Análise de estrutura: distância, ângulos, ligações de hidrogênio e sobreposição (matchmaker)



Ferramentas do ChimeraX

Principais ferramentas:

- Edição de estrutura:
Rotâmeros, alterar chain ID e numeração de residues



Mais ferramentas

ChimeraX Toolshed - All Apps

https://cxtoolshed.rbvi.ucsf.edu/apps/all?platform=Linux&version=1.10

search in page

All Bundles

Categories

EM

Structure Analysis

Input/Output

User Interface

Structure Editing

atomic structure

MD

Fitting

external program

Protein Visualization

Model validation

crystallography

augmented reality

Sequence

Light microscopy

ligands

developer

Cytoscape

presets

NMR

more »

Get Started with the Toolshed »



AddH

Add hydrogens



AllMetal3D

Predicting metal and water binding sites in proteins



ArtiaX

Visualization and editing of cryo-ET particle lists.



Clipper

Adds support for crystallographic maps to



Cytoscape

This bundle provides an interface to Cytoscape's



DevelExtras

Work with non installed Bundles



DiffFit

Rapid Fitting of Molecular Structures to a Cryo-EM



EMalign

Fully automatic alignment of density maps



GenomeTools

Work with 3D genome models



HKCage

Depict icosahedral lattices for viruses



ISOLDE

Interactive Structure Optimization by Local Direct



LAMMPS

Import .data(.gz) and .dump(.gz) from LAMMPS



LeapMotion

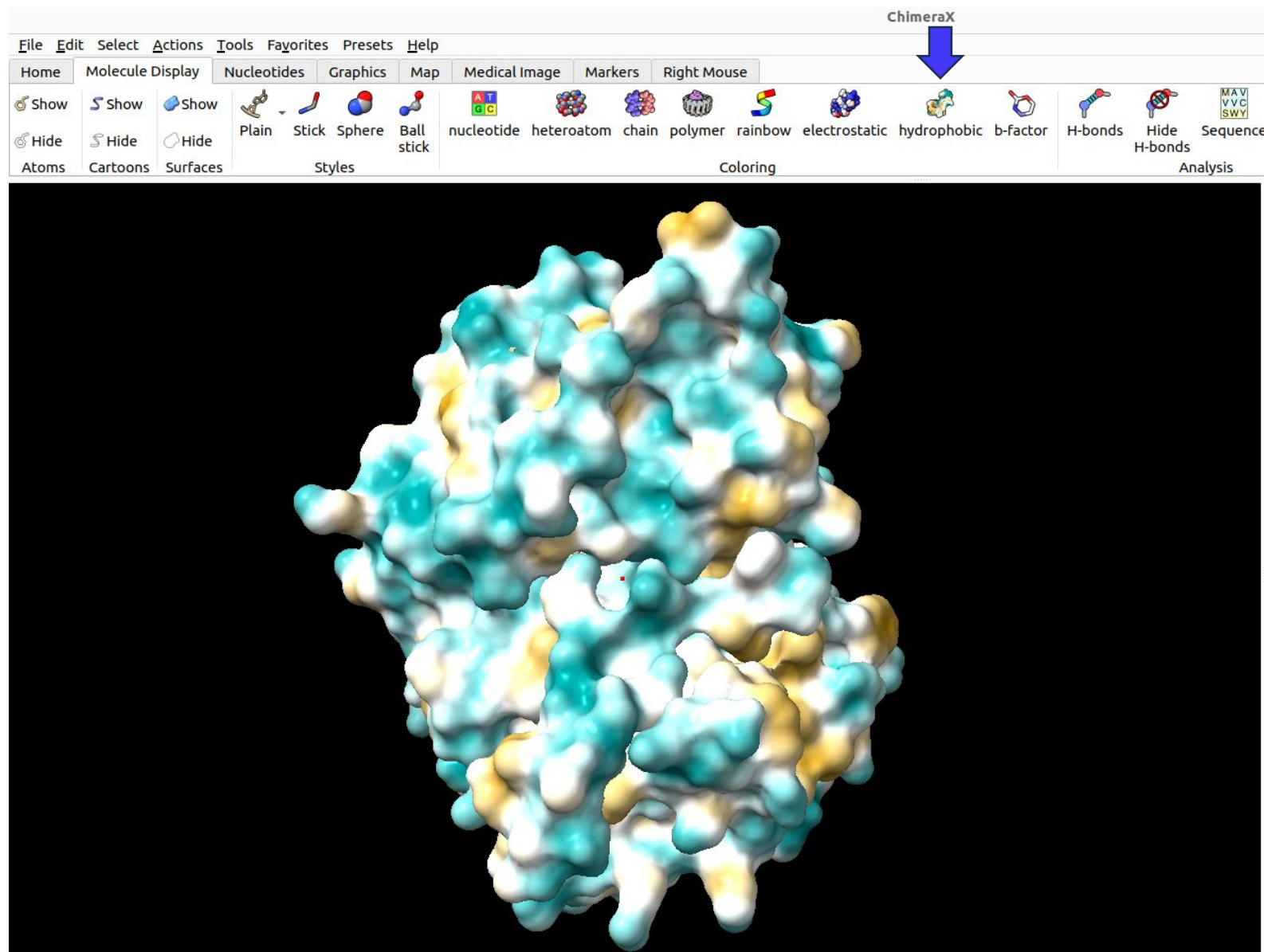
Leap Motion hand tracking



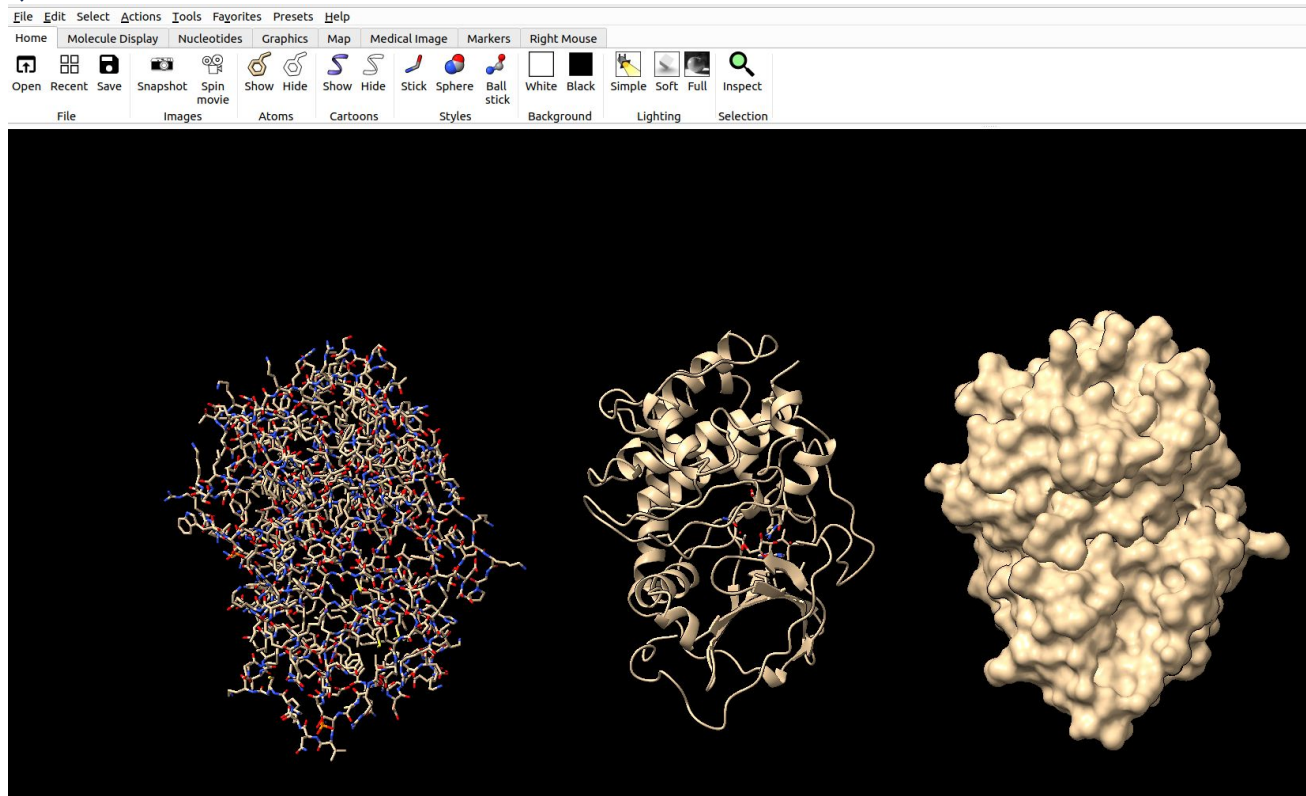
LigandRecognizer

Commands for recognizing ligands in cryoEM maps

Gerando representação de superfície colorida por hidrofobicidade



Salvando imagens no ChimeraX



File name:

Files of type: PNG image (*.png)

Size: 1269 x 804 ☒ preserve aspect Supersample: 3x ☐ Transparent background

Generating videos with ChimeraX

- **2dlabels** – add text, [symbols](#), and straight arrows to the display for presentation images and movies
- **camera** – set mono or stereo viewing and related parameters
- **clip** – control clipping planes
- **cofr** – set center of rotation
- **coordset** – play through frames of a [trajectory](#)
- **crossfade** – interpolate between image frames
- **fly** – smoothly traverse a series of named views
- **morph** – morph (interpolate) between atomic structures; create a morph trajectory
- **move** – translate
- **movie** – record image frames and assemble them into a movie file
- **mseries** – display an [ordered series of models](#)
- **perframe** – specify operations to execute at every (or every Nth) display frame
- **resfit** – show density fit of successive amino acid residues along a chain
- **rock** – rock back and forth
- **roll** – rotate continuously
- **scenes** – save/restore named scenes
- **stop** – halt ongoing motions
- **torsion** – set or report torsion angles (rotate bonds)
- **turn** – rotate
- **view** – adjust the view to specified items; save/restore named views
- **volume morph** – morph (interpolate) between two or more maps
- **vr roomCamera** – in [virtual reality](#) (VR), set up a separate camera view fixed in the VR room coordinates
- **vseries** – display, analyze, or process an [ordered series of maps](#)
- **wait** – update the display and enforce ordered execution of commands in scripts
- **window size** – set pixel width and height of the graphics window
- **wobble** – perform a figure-eight rotation
- **zoom** – change the apparent size of the view
- **record** – start recording frames
- **encode** – stop any ongoing recording and encode the saved image frames into a movie file
- **stop** – stop recording frames
- **reset** – reset status to zero frames saved
- **abort** – halt any encoding in progress and perform a **reset**
- **crossfade** – interpolate from the preceding frame to the following frame
- **duplicate** – expand the preceding frame into multiple frames
- **ignore** – start/stop ignoring subsequent **movie** commands
- **formats** – list the available movie file formats
- **status** – report status (number of saved frames, *etc.*) in the [Log](#)

Obrigado!

Equipe de Dados Biológicos (EDB)

edb@lnbio.cnpem.br

+55 (19) 3512-1113

Grupo de Design de Proteínas (GDP)

+55 (19) 3512-2389

Laboratório de Biologia Computacional (LBC)

+55 (19) 3512-1255



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